The ethics of generative AI in software engineering

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1 Introduction

Software engineering is a field that has developed most of today's infrastructure in some way or another, be it via the implementation of a medical device to the websites that allow us to connect the world. However, these projects need to manage several stakeholders, engineers and budgets to make things come to reality. With the improvements in generative AI (GenAI for short), there has been a shift towards trying to incorporate this new tool into the software development pipeline. These attempts are not small companies but instead industry leaders such as Microsoft. [1]

Given the workings of generative AI, there are several key concerns: The displacement of people for compute centers, the reliability of the software generated and the dilution of understanding the software developed. This essay will try to answer the question:

"Is it ethical for software engineers to use modern generative AI applications to generate code that is used, in whole or in part, for commercial applications?"

2 Core Issues

As discussed in the introduction, there are several concerns with the use of generative AI in software development:

2.1 Software reliability

One of the major issues that stems from using GenAI is the fact that GenAI doesn't *reason* like humans do. Unlike humans that

2.2 Dilution of knowledge

2.3 Displacement of populations

3 Stakeholders

- General population
- Software engineers
- Companies that use generative AI
- Companies that make generative AI tools

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4 Current state of generative AI

This section will focus on the core applications and issues that generative AI is undergoing:

Applications:

- Usage in research
- Vibe coding
- Usage in open source projects
- Usage in proprietary projects
- Data privacy

Issues:

- The ethics of training it on codebases without the creator's explicit approval
- Hallucinations: The system's inability to tell you that it doesn't know something
- The "bubble": The financial risks associated with running such an expensive business
- Population displacement: The space and resources needed to have the compute centers
- Unreliability: Leaks and the such

5 Technical Background

The technical section will focus on the following:

- The math of the models
- LLMs and how they work
- Code generation models and what extra features they implement to ensure correct code
- Why hallucinations occur

6 Ethical Analysis

6.1 Kantianism

Under Kantianism, the biggest ethical concerns are:

- Using code for training without the owner's conscent (GitHub Copilot)
- Displacing people to fit more compute centers
- Using more water to keep compute centers cool

6.2 Act Utilitarianism

In act utilitarianism the biggest contrast is whether all the negative effects on other humans helps maximize the happiness in other humans. This can be evaluated through the development of new technologies and research. By evaluating different sources it is possible to evaluate such cases.

6.3 Virtue Ethics

Under virtue ethics, the focus is the virtues of the stakeholders and how the decisions they make with this technology and how virtues either align or misalign with them.

6.4 Social Contract Theory

This section focuses on what our society expects out of software and how the usage of machine learning can breach the agreements and expectations that we have about the software that we use and how it is made.

7 Conclusion

8 Beyond the surface

References

[1] Jaime Teevan, Sonia Jaffe, Rebecca Janssen, Nancy Baym, Siân Lindley, Bahar Sarrafzadeh, Brent Hecht, Jenna Butler, Jake Hofman, and Sean Rintel. [n. d.]. *New Future of Work: AI is driving rapid change, uneven benefits.* https://www.microsoft.com/en-us/research/blog/new-future-of-work-ai-is-driving-rapid-change-uneven-benefits/